Superconductivity: Materials and Underlying Mechanisms

By Oleksandr Boyko (Alex)

Abstract

This article explains the phenomenon of superconductivity as well as fundamental properties of superconductors. It examines the quantum mechanical mechanisms that enable electrical current to flow within the superconductor without resistance, as well as the expulsion of magnetic fields, by a superconducting material, known as the Meissner effect. In addition the article outlines the history of invention and improvement of superconducting materials, and their real world applications.

Introduction

Metals are widely used as electrical conductors due to their ability to allow the flow of electric current with relatively low resistance. The conductivity of the material is inversely proportional to its resistivity. However, there are certain materials that allow current to flow without energy loss, resulting in effectively zero electrical resistance. These materials are known as superconductors.

1. What are superconductors?

Superconductors are materials whose electrical resistance drops to zero when they are cooled below their transition temperature (the specific, often extremely low, temperature at which a material transitions from a normal state to a zero-resistance superconducting state)[1]. There are a number of metallic elements (aluminium, lead, mercury) that exhibit superconductivity if cooled sufficiently, although their transition temperature is extremely low (<10 K). This makes them unsuitable for reliable and practical use due to the logistical and monetary issues that arise from the necessity of obtaining such low temperatures.

As a result of extensive research in the field of superconductivity, some materials with a much higher transition temperature were discovered. Notably, ceramic compounds such as yttrium barium copper oxide (YBCO) and bismuth strontium calcium copper oxide (BSCCO) exhibit superconductivity at temperatures above 93 K and 108 K respectively[2]. These so-called high-temperature superconductors have significantly increased the feasibility of practical applications, particularly in areas where liquid nitrogen cooling can be used.

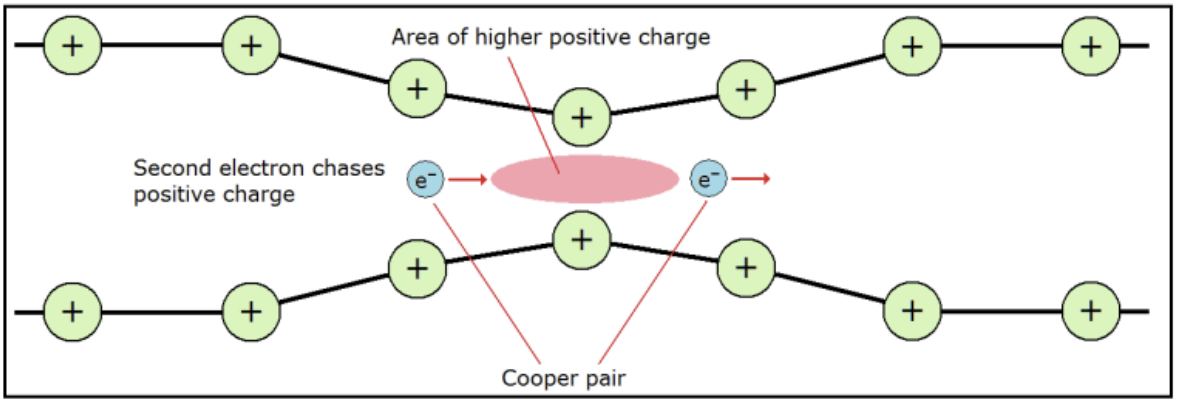
2. Mechanism of Superconductivity

I. Zero resistance below their transition temperature

In conventional conductors, electrical resistance decreases gradually as temperature is reduced, primarily due to reduced lattice vibrations and fewer collisions between the passing electrons and atomic nuclei. This implies that the resistivity of materials decreases gradually with the decrease in temperature[3]. However, this classic behaviour cannot account for superconductivity, where resistance drops abruptly to zero at a transition temperature.

While electrons behave as particles, they undergo scattering from lattice imperfections and collisions with atoms, which means that for non-superconductors the resistivity will never reach zero. However, once a superconductor reaches its transition temperature, the phenomenon known as Cooper Pair formation occurs. When the electron travels through the lattice of positively charged ions, it induces a slight distortion of the lattice structure, creating a region of increased positive charge density. Under sufficiently low temperatures and appropriate lattice structure, this area of positive charge attracts another electron and, as a result, a pair of electrons is formed, which is called a Cooper Pair[4] (*Figure 1.1*).

Figure 1.1 The simplified model of formation of Cooper Pairs



https://dc.edu.au/wp-content/uploads/cooper-pair-phonon.png

Source: PHYS 498 ART Spring 2018 by Karmela Padavić

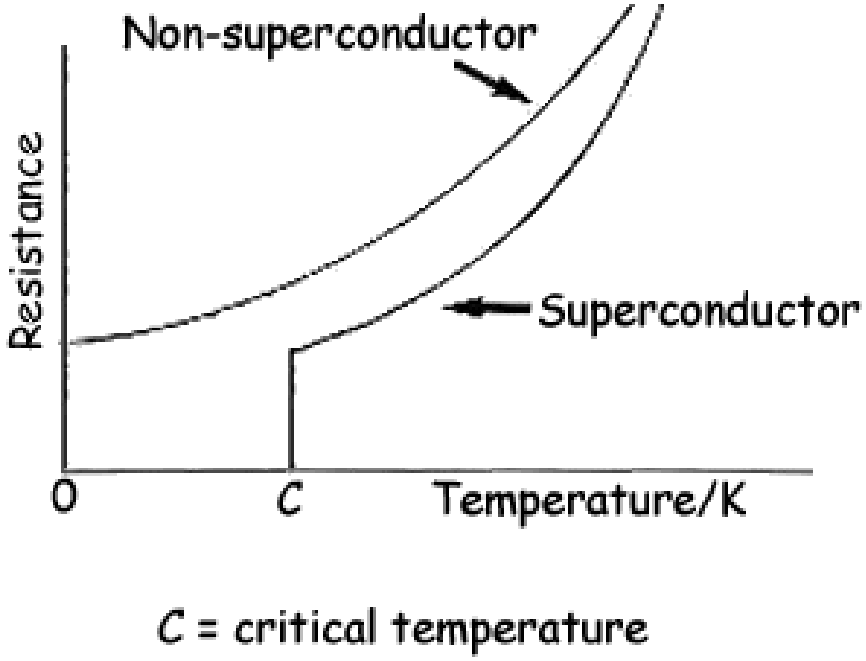
Under normal conditions, the electron is a subatomic particle classified as a fermion, which means it has a half-integral spin (a quantum mechanical property of particles where their intrinsic angular momentum is a half-integer multiple of the reduced Planck constant), obeying the Pauli Exclusion Principle (no two electrons can have the same four electronic quantum numbers, meaning they cannot exist at the same place at the same time), but once 2 electrons form a pair their spins align in the opposite direction to one another, resulting in the net spin of 0. This means that 2 electrons in a Cooper Pair behave like a composite boson rather than a fermion. Bosons are not subject to the Pauli Exclusion Principle, allowing them to occupy the same quantum state (similarly to how photons can occupy the same quantum space in a laser), forming a wave described by a single macroscopic wavefunction[4].

This coherent quantum state allows electrons to move through the material without scattering, eliminating electrical resistance. Unlike in normal conductors, where electrons behave as independent particles, superconducting electrons act collectively, preventing energy dissipation. As a result, the wavefunction they abide by is robust to the scattering

from the atoms or imperfections in the substance, effectively creating a current with 0 resistance.

This explains why in conventional conductors resistivity gradually decreases with the decrease of the temperature, whereas in superconductors it drops abruptly to zero at the transition temperature (*Figure 1.2*)

Figure 1.2 The relationship between the resistance and temperature for the superconductors and non-superconductors.



Source: cyberphysics

II. No interior magnetic field

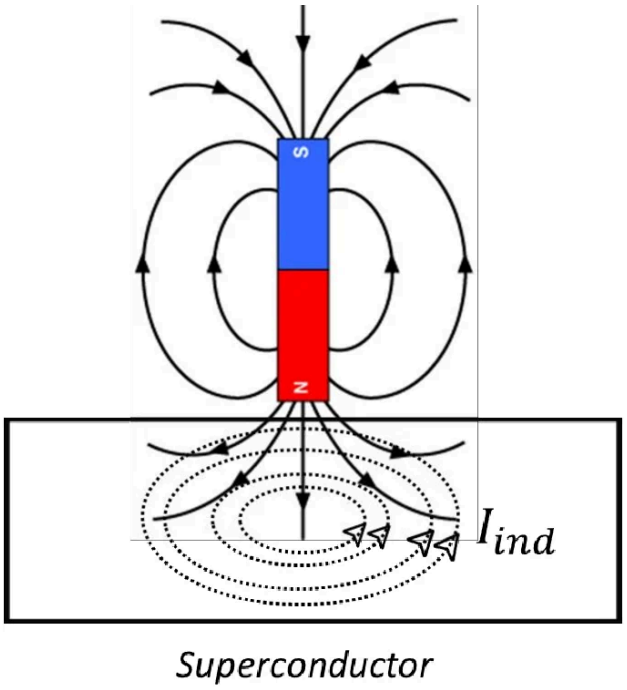
Another key property of all superconductors is the complete expulsion of external magnetic fields. This property arises as a result of a unique superconducting state, rather than simply the absence of electrical resistance. The cause of this phenomenon can be explained by The Meissner effect.

As stated above superconductors do not have resistance thus once there is a current in the circuit made entirely of a superconducting material it theoretically will never diminish unless the external force acts upon it. Once the superconductor is introduced to a magnetic field, persistent surface currents are induced within the material - eddy currents (a localized electric current induced in a conductor by a varying magnetic field) (Figure 2.1). Since such currents are not affected by resistance within the superconductor their magnitude is directly proportional to the strength of the magnetic field to which they are exposed. However, those currents also produce their own magnetic field. This field adjusts in response to the applied magnetic field, opposing and cancelling it within the interior of the superconductor. This means that the magnetic field produced by the current in the superconductor is equal in

magnitude and mirrored (opposite) in direction to the external magnetic field that acts upon the superconductor, effectively counteracting it.

Moreover, unlike a conventional conductor, a superconductor will expel a magnetic field even if that field is already present when the material is cooled below its critical temperature. This demonstrates that the effect is not simply due to electromagnetic induction, but is instead a fundamental property of the superconducting state.

Figure 2.1 Diagram, showing the creation of eddy currents induced by the external magnetic field



Source: Researchgate 2022 by Nabin Raut

This phenomenon explains the absence of an internal magnetic field in the superconductor as well as why a magnet placed above a superconductor can levitate in a stable position.

However, this effect has its limitations. For each superconducting material there is a specific strength of magnetic field known as a critical magnetic field, beyond which the current within the superconductor cannot rise anymore and superconductivity is destroyed, with the material returning to its normal resistive state.

3. History of superconductors

The discovery of superconductors was accidental. In 1911 the Dutch physicist Heike Kamerlingh Onnes in his laboratory noticed mercury's zero resistance at temperatures lower than 4.2 K. It has led to the creation of an entirely new field within physics called superconductivity. Since then the physics behind superconductors is rigorously studied leading to the discovery of new materials and methods of their synthesis.

Their previously unseen properties have puzzled physicists all around the world, and many explanations of the phenomena were proposed with little to no success. Until in 1935, the theory proposed by Fritz and Heinz London (known as the London theory) was the first success in the field, which was followed by another theory put forward by Vitaly Ginzburg and Lev Landau in the 1950s.

Though the proposed theory was satisfactory it was not perfect and soon a more complete explanation emerged with the work of John Bardeen, Leon Cooper and John Robert Schrieffer. It was nicknamed the BCS theory in their honour. This theory successfully explained superconductivity in conventional materials through the formation of Cooper pairs and remains fundamental to the field.

Later another breakthrough in physics occurred - the discovery of the cuprate high temperature superconductors (HTS) in 1986, which gave a new momentum to the field and intensified the research towards the development of superconductors that can operate at even higher, potentially room, temperatures, which continues to this day.

4. Real life applications of superconductors

Superconductors already have a range of practical applications, as cooling materials to the boiling points of liquid helium or hydrogen is technically feasible. However, conditions of their use are quite demanding in the face of high costs and complicated logistics used to maintain sufficient temperatures.

Superconducting materials have been explored for use in high-speed connections between components inside high-level computer chips, where reduced electrical resistance can significantly improve performance. In medicine, superconducting coils enabled the generation of extremely strong magnetic fields in devices such as magnetic resonance imaging (MRI) scanners, which are used to produce detailed images of soft tissues within the human body, improving treatments that involve precise surgeries like brain tumor removal.

In countries like Japan and Germany, superconducting magnets are used in experimental magnetic levitation (maglev) trains, allowing them to hover above the tracks and therefore minimise friction, improving efficiency and speed. Similarly, superconducting magnets play a crucial role in particle accelerators, where they are used to guide and focus particle beams for the study of fundamental atomic and subatomic structures.

Bibliography

[1]

K. Bhaumik, “Superconductivity: A Review,” *Researchgate*, Jan. 05, 2022. http://www.researchgate.net/publication/357604575_Superconductivity_A_Review

[2]

D. Homa, G. Pickrell, and G. Kaur, “On the verge of high temperature superconducting fibers,” *Materials Letters*, vol. 121, pp. 101–104, Feb. 2014, doi: https://doi.org/10.1016/j.matlet.2014.01.147.

[3]

R. A. Butera and D. H. Waldeck, “The Dependence of Resistance on Temperature for Metals, Semiconductors, and Superconductors,” *Journal of Chemical Education*, vol. 74, no. 9, p. 1090, Sep. 1997, doi: https://doi.org/10.1021/ed074p1090.

[4]

J. Bardeen, L. N. Cooper, and J. R. Schrieffer, “Theory of Superconductivity,” *Physical Review*, vol. 108, no. 5, pp. 1175–1204, Dec. 1957, doi: https://doi.org/10.1103/physrev.108.1175.

[5]

W. J. Mercer and Y. A. Pashkin, “Superconductivity: the path of least resistance to the future,” *Contemporary Physics*, pp. 1–28, Nov. 2023, doi: https://doi.org/10.1080/00107514.2023.2259654.